

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276579

Kind Code

A1

Publication Date

September 04, 2025

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VEHICLE AUDIO CONTROL DEVICE

Abstract

A vehicle audio control device includes an electronic controller, a computer-readable medium and an audiovisual system. The electronic controller is programmed to be in communication with a sensor system of an electrified vehicle to receive signals from the sensor system regarding a state of the electrified vehicle during a vehicle drive cycle. The computer-readable medium stores a plurality sound outputs. Each of the sound outputs corresponds to one or more states of the vehicle drive cycle. The user interphase is operable to set the sound outputs with respect to the one or more states of the vehicle drive cycle. The audiovisual system has at least one speaker controlled by the electronic controller to generate the sound outputs when the electrified vehicle is in the corresponding state of the vehicle drive cycle.

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Family ID: 96881790

Appl. No.: 18/592015

Filed: February 29, 2024

Publication Classification

Int. Cl.: B60K35/26 (20240101); B60K35/21 (20240101)

U.S. Cl.:

CPC B60K35/26 (20240101); B60K35/213 (20240101); B60K2360/151 (20240101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to a vehicle audio control device. More specifically, the present disclosure relates to a vehicle audio control device that allows for customized sound selections during vehicle driving.

BACKGROUND INFORMATION

[0002] Electrified vehicles use of one or more electric motors alone or in combination with an internal combustion engine to power the vehicle and often include transmissions or gear boxes such as a continuously variable transmission or power split arrangement that do not include discreet gear ratios associated with conventional manual or automatic transmissions. Active sound designs can be implemented for electric vehicles to create consistent acoustic feedback for the driver, reflecting driving conditions (acceleration, braking, etc.), which is absent in electrified vehicles.

SUMMARY

[0003] In view of the state of the known technology, an aspect of the present disclosure is to provide a vehicle audio control device comprising an electronic controller, a computer-readable medium and an audiovisual system. The electronic controller is programmed to be in communication with a sensor system of an electrified vehicle to receive signals from the sensor system regarding a state of the electrified vehicle during a vehicle drive cycle. The computer-readable medium stores a plurality sound outputs. Each of the sound outputs corresponds to one or more states of the vehicle drive cycle. The user interphase is operable to set the sound outputs with respect to the one or more states of the vehicle drive cycle. The audiovisual system has at least one speaker controlled by the electronic controller to generate the sound outputs when the electrified vehicle is in the corresponding state of the vehicle drive cycle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Referring now to the attached drawings which form a part of this original disclosure:

[0005] FIG. **1** is a schematic overview of a vehicle having a vehicle audio control device implemented therein in accordance with an illustrated embodiment;

[0006] FIG. **2** is a schematic view of the vehicle audio control device;

[0007] FIG. **3** is a sample of information displayed for the vehicle audio control device;

[0008] FIG. **4** is a sample vehicle drive cycle for a vehicle having the vehicle audio control device implemented therein; and

[0009] FIG. **5** is another sample vehicle drive cycle for another vehicle having the vehicle audio control device implemented therein.

DETAILED DESCRIPTION OF EMBODIMENTS

[0010] Selected embodiments will now be explained with reference to the drawings. It will be apparent to those skilled in the art from this disclosure that the following descriptions of the embodiments are provided for illustration only and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

[0011] Referring initially to FIGS. **1** and **2**, a vehicle audio control device **10** is illustrated in accordance with an embodiment. The vehicle audio control device **10** can be provided or implemented with an electrified vehicle **12**. In the illustrated embodiment, the vehicle audio control device **10** preferably further comprises the electrified vehicle **12**. The electrified vehicle **12** can be an electric vehicle (EV) or a hybrid electric vehicle (HEV). That is, the electrified vehicle **12** can be a battery operated electric vehicle powered only by a battery and one or more electric machines that may be operated as a motor/generator selectively drawing power from or charging a high voltage battery (not shown). Alternatively, the vehicle audio control device **10** can be implemented with an HEV powered by an internal combustion engine in combination with one or more electric

machines that may each operate as a motor and/or generator. The vehicle audio control device **10** can be implemented with various types of passenger vehicles, such as crossover utility vehicle (CUV or XUV), sport utility vehicle (SUV), truck, recreational vehicle (RV), etc. including driver operated, driver-assisted, and fully autonomous configurations for transporting people or goods. [0012] Additionally, the vehicle audio control device **10** further comprises an electronic controller **14** and a computer-readable medium **16** that can control an audiovisual system **18** of the vehicle. The audiovisual system **18** includes at least one user interphase **20** and a speaker system **22** of the electrified vehicle **12**. The audiovisual system **18** preferably further includes at least one electronic display **24**. The speaker system **22** includes at least one speaker of the electrified vehicle **12**. The electronic controller **14** is in communication with a sensor system **26** of the electrified vehicle **12** to receive information regarding a state of the electrified vehicle **12** during a vehicle drive cycle as will be further explained.

[0013] The electronic controller **14** further controls the user interphase **20**, the electronic display **24**, and the speaker system **22** of the electrified vehicle **12** in accordance with the current vehicle drive cycle. In particular, the vehicle audio control device **10** enables the speaker system **22** to play sounds corresponding to pre-selected sounds during different events of a vehicle drive cycle as will be further described below. Thus, the vehicle audio control device **10** allows for customizable sound generation in the electrified vehicle **12** throughout different states of the electrified vehicle's **12** drive cycle. The term "drive cycle" as used herein refers to a series of driving states or events of the electric vehicle **12** during instances of use. Each drive cycle can vary and include different states or events of the vehicle **12** as th text missing or illegible when filed

[0014] The vehicle **12** undergoes different behaviors under different circumstances and encounters. The at least one electronic display **24** that is preferably positioned in an interior compartment of the vehicle **12** as part of the console. The electronic display **24** can also be a heads-up display on the vehicle's **12** windshield. Alternatively, the electronic display **24** can include or be a mobile device that is in electronic communication with the electronic controller **14** to display information regarding the audiovisual system **18**. The display **24** can include a single type display, or multiple display types (e.g., both audio and visual) configured for human-machine interaction. The electronic display **24** displays different information regarding the sound outputs for the vehicle audio control device **10**, as seen in FIG. 3.

[0015] The user interphase **20** is operable to set the sound outputs with respect to the one or more states of the vehicle drive cycle. Therefore, the electronic controller **14** can receive user inputs from the vehicle **12** occupants via the user interphase **20** to set the desired sound output as will be described below. The user interphase **20** can be part of or integral with the electronic display **24** which can include, for example, control buttons and/or control buttons displayed on a touchscreen display which enable the user to enter commands and information, as seen in FIG. 3. Alternatively, the user interphase **20** can be provided as part of the steering wheel audio buttons, a push-to-talk button (e.g., hard buttons and/or soft buttons), instrument panel controls, etc. Alternatively, the user interphase **20** can be provided on a mobile device that is in communication with the electronic controller **14**.

[0016] The electronic controller **14** is in communication with the at least one speaker of the audiovisual system **18** to provide sound output to vehicle occupants. The vehicle audio control device **10** is provided as an active sound system to create consistent acoustic feedback for the driver, reflecting driving conditions (acceleration, braking, etc.), which is absent in electrified vehicles that lack powertrain noise. The vehicle audio control device **10** can control interior speakers of the electrified vehicle **12** for the passenger to compensate for the lack of acoustic feedback and increase driving pleasure. Secondly, new noise sources that have become apparent can be partially masked by enhancing the sound and providing the right driving experience. Therefore, the vehicle audio control device **10** enables driving the electrified vehicle **12** to have the perception of driving a sporty car just by playing the right active sound over the speaker system **22**.

Additionally, the vehicle audio control device **10** of the illustrated embodiment can control exterior speakers of the electrified vehicle **12** to generate sounds for exterior vehicles in the vicinity of the electrified vehicle **12** that can help the electrified vehicle **12** meet current legal requirement of acoustic vehicle alerting systems (AVAS) in electrified vehicle **12s** to achieve minimal noise levels. [0017] As stated, the electronic controller **14** is programmed to be in communication with the sensor system **26** to receive signals from the sensor system **26** regarding a state of the electrified vehicle **12** during a vehicle drive cycle. In the illustrated embodiment, the sensor system **26** can include a controller area network (CAN-bus) system that is a message-based protocol designed to allow the electronic control units (ECUs) found in automobiles, as well as other devices, to communicate with each other in a reliable, priority-driven fashion.

[0018] Various sensor inputs from around the vehicle (speed sensors, steering angle, air conditioning on/off, engine temperature) are collated via the CAN-bus, as illustrated in FIG. **1**. For example, the electronic controller **14** can monitor signals from the CAN-bus to receive signals from the accelerator pedal sensor to monitor a degree of accelerator opening, the brake pressure sensor to determine deceleration, the engine/motor rotation sensor to determine vehicle speed, etc. Therefore, the electronic controller **14** can receive signals regarding the accelerator pedal position, the vehicle speed, the motor speed, the motor torque, and/or sensed vibrations. Therefore, the signals from the sensor system **26** provide information regarding a state of the drive cycle of the vehicle **10** such as acceleration, deceleration, coasting, sport mode, etc. Additionally, the electronic controller **14** can receive signals from the start/stop system of the electrified vehicle **12** if the electrified vehicle **12** is an HEV to determine whether the vehicle **12** is at a stop or stationary. In the illustrated embodiment, the sound outputs are active sounds created by real-time sound synthesis based on the CAN-bus input signals.

[0019] The electronic controller **14** can also be in communication with the vehicle's **12** electric park brakes, the park assist systems, auto lane assist/collision avoidance systems to determine whether the vehicle is parking, changing lanes, turning, etc. It will be apparent to those skilled in the vehicle field from this disclosure that the vehicle audio control device **10** can be in communication with other types of sensors and positioning systems of the vehicle (e.g., cameras, bumper sensors, radar, antennas, the global positioning unit, telemetric, etc.) to control the speaker system **22** to generate different sounds according to different information received from these systems. Therefore, the electronic controller **14** is programmed to be in communication with the sensor system **26** to receive signals from the sensor system **26** regarding a state of the electrified vehicle **12** during a vehicle drive cycle.

[0020] As stated, the electronic controller **14** controls the audiovisual system **18** in accordance with the drive cycle. That is, the electronic controller **14** can control the speakers to broadcast sounds through the speakers based on the current state of the vehicle **12** in the drive cycle. For example, when the state of the drive cycle is that the vehicle **12** is in park or neutral the electronic controller **14** can control the speakers to play a desired sound, such as an iconic engine idle sound. When the vehicle **12** then accelerates, the electronic controller **14** can determine that the state of the drive cycle is that the vehicle **12** is accelerating based on the accelerator pedal position and control the speakers to generate a desired sound such as revving of the engine. Therefore, the sound generated by the speakers transitions from a neutral sound to a revving sound as the vehicle **12** transitions between different states in the drive cycle.

[0021] In the illustrated embodiment, the sound outputs are customizable via the user interphase **20** that is operable by the user to set the sound outputs with respect to the one or more states of the vehicle drive cycle. The selections for the sound outputs are displayed on the electronic display **24**. Thus, the electronic display **24** is programmed to display a selection of the sound outputs with respect to the one or more states of the vehicle drive cycle. The user can select a desired sound output (e.g., idling sound) for when the vehicle **12** is at an idle state. The user can select another desired sound output for when the vehicle **12** is at an acceleration state (e.g., revving sound) and

when the vehicle **12** is at a cruising state (e.g., desired music). Therefore, the speakers are controlled by the electronic controller **14** to generate the sound outputs when the electrified vehicle **12** is in the corresponding state of the vehicle drive cycle.

[0022] The state of the drive cycle can include, but is not limited to, an ON/OFF event, an acceleration state event, a turning state event, a lane changing event, a cruising event, a deceleration event, a parking event, a stop event, a cruising event. Therefore, the state of the electrified vehicle drive cycle includes any one or more of a vehicle ON event, a vehicle OFF event, a start event, a stop event and a vehicle drive event. In the illustrated embodiment, the vehicle drive event includes any one or more of an acceleration event, a deceleration event, a reverse event, a turning event, a lane change event, a cruising event and an idle event. The vehicle audio control device **10** of the illustrated embodiment allows the user to control when they hear the desired sound output. For example, the user can set the vehicle control device to play only a desired sound during an acceleration event but not while cruising at constant speed on highway. The user can also customize sound during a deceleration event to either play a desired sound output or to be entirely OFF.

[0023] For example, as seen in FIG. **4**, the vehicle **V1** is an example of a vehicle that is provided with the vehicle audio control device **10**. The vehicle **V1** is decelerating at **P1** to avoid the cyclist, the vehicle **V1** is undergoing a lane change at **P2**, accelerating after the lane change at **P3** and then cruising at **P4**. Therefore, the vehicle **V1** is undergoing a deceleration event, a lane change event, an acceleration event and a cruising event. The vehicle audio control device **10** can be set to not play any sounds during lane changes or deceleration events (at **P1** or **P2**), but to play a desired sound or music during the acceleration event and the cruising event (at **P3** and **P4**). Therefore, the electronic controller **14** can control the speakers to be silent during deceleration and lane change but to resume sounds during acceleration and cruising.

[0024] Referring to FIG. **5** in another example, the vehicle **V2** is another example of a vehicle that is provided with the vehicle audio control device **10**. The vehicle **V2** is decelerating at **P100**, idling at **P200** and parking at **P300**. The vehicle audio control device **10** can control the speakers to generate different sound outputs or be entirely OFF at any of these changing states of the vehicle drive cycle in accordance with user customization. Therefore, the electronic controller **14** is programmed to control the at least one speaker to change the sound output that is generated based on a change in the state of the electric vehicle drive cycle.

[0025] The electronic controller **14** includes a processor or CPU that controls the operation of the vehicle control apparatus. As used herein, the terminology “processor” indicates one or more processors, such as one or more special purpose processors, one or more digital signal processors, one or more microprocessors, one or more controllers, one or more microcontrollers, one or more application processors, one or more Application Specific Integrated Circuits, one or more Application Specific Standard Products; one or more Field Programmable Gate Arrays, any other type or combination of integrated circuits, one or more state machines, or any combination thereof.

[0026] The electronic controller **14** further has a computer readable medium storing the sound output options. As used herein, the terminology “memory” or “computer-readable medium **16**” (also referred to as a processor-readable medium) indicates any computer-usable or computer-readable medium **16** or device that can tangibly contain, store, communicate, or transport any signal or information that may be used by or in connection with any processor. For example, the computer-readable medium **16** may be one or more read only memories (ROM), one or more random access memories (RAM), one or more registers, low power double data rate (LPDDR) memories, one or more cache memories, one or more semiconductor computer readable medium devices, one or more magnetic media, one or more optical media, one or more magneto-optical media, or any combination thereof.

[0027] Therefore, the computer-readable medium **16** further includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a

computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Non-volatile media can include, for example, optical or magnetic disks and other persistent computer readable medium. Volatile media may include, for example, dynamic random access computer readable medium (DRAM), which typically constitutes a main computer readable medium.

[0028] The computer-readable medium **16** can also be provided in the form of one or more solid state drives, one or more computer readable medium cards, one or more removable media, one or more read-only memories, one or more random access memories, one or more disks, including a hard disk, a floppy disk, an optical disk, a magnetic or optical card, or any type of non-transitory media suitable for storing electronic information, or any combination thereof.

[0029] The computer-readable medium **16** comprises a plurality sound outputs, each of the sound outputs corresponding to one or more states of the vehicle drive cycle. As seen in FIG. 3, the electronic display **24** can display options for the sound outputs that is stored in the computer-readable medium **16**. In the illustrated example, the sound outputs are depicted as “sound output selection” on the electronic display **24**, with each sound output corresponding to different driving states or events so to be played at those different driving states or events. The sound outputs can be prestored in the computer-readable medium **16** or stored on a cloud for which the user has access via a subscription service. Therefore, the sound outputs can be provided via over-the-air (OTA) updates or a subscription service.

[0030] The electronic controller **14** can be further programmed create a plurality of sound profiles, each of the sound profiles corresponding with one or more states of the vehicle drive cycle. Therefore, each state of the drive cycle has a sound profile which includes customized settings for ON/OFF, sound selection, sound volume, and/or sound responses for the specific drive state or event.

[0031] The processor of the electronic controller **14** can execute instructions transmitted by one or more transmission media, including coaxial cables, copper wire and fiber optics, including the wires that comprise a system bus coupled to a processor of a computer. As used herein, the terminology “instructions” may include directions or expressions for performing any method, or any portion or portions thereof, disclosed herein, and may be realized in hardware, software, or any combination thereof.

[0032] For example, instructions may be implemented as information, such as a computer program, stored in computer readable medium that may be executed by a processor to perform any of the respective methods, algorithms, aspects, or combinations thereof, as described herein. In some embodiments, instructions, or a portion thereof, may be implemented as a special purpose processor, or circuitry, that may include specialized hardware for carrying out any of the methods, algorithms, aspects, or combinations thereof, as described herein. In some implementations, portions of the instructions may be distributed across multiple processors on a single device, on multiple devices, which may communicate directly or across a network such as a local area network, a wide area network, the Internet, or a combination thereof.

[0033] Computer-executable instructions can be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Visual Basic, Java Script, Perl, etc. In general, the processor receives instructions from the computer-readable medium **16** and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer-readable media.

[0034] In understanding the scope of the present invention, the term “comprising” and its derivatives, as used herein, are intended to be open ended terms that specify the presence of the stated features, elements, components, groups, integers, and/or steps, but do not exclude the presence of other unstated features, elements, components, groups, integers and/or steps. The

foregoing also applies to words having similar meanings such as the terms, “including”, “having” and their derivatives. Also, the terms “part,” “section,” “portion,” “member” or “element” when used in the singular can have the dual meaning of a single part or a plurality of parts.

[0035] The term “detect” as used herein to describe an operation or function carried out by a component, a section, a device or the like includes a component, a section, a device or the like that does not require physical detection, but rather includes determining, measuring, modeling, predicting or computing or the like to carry out the operation or function.

[0036] The term “configured” as used herein to describe a component, section or part of a device includes hardware and/or software that is constructed and/or programmed to carry out the desired function.

[0037] The terms of degree such as “substantially”, “about” and “approximately” as used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed.

[0038] While only selected embodiments have been chosen to illustrate the present invention, it will be apparent to those skilled in the art from this disclosure that various changes and modifications can be made herein without departing from the scope of the invention as defined in the appended claims. For example, the size, shape, location or orientation of the various components can be changed as needed and/or desired. Components that are shown directly connected or contacting each other can have intermediate structures disposed between them. The functions of one element can be performed by two, and vice versa. The structures and functions of one embodiment can be adopted in another embodiment. It is not necessary for all advantages to be present in a particular embodiment at the same time. Every feature which is unique from the prior art, alone or in combination with other features, also should be considered a separate description of further inventions by the applicant, including the structural and/or functional concepts embodied by such feature(s). Thus, the foregoing descriptions of the embodiments according to the present invention are provided for illustration only, and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

Claims

1. A vehicle audio control device comprising: an electronic controller programmed to be in communication with a sensor system of an electrified vehicle to receive signals from the sensor system regarding a state of the electrified vehicle during a vehicle drive cycle; a computer-readable medium comprising a plurality sound outputs, each of the sound outputs corresponding to one or more states of the vehicle drive cycle; a user interphase operable to set the sound outputs with respect to the one or more states of the vehicle drive cycle; and an audiovisual system having at least one speaker controlled by the electronic controller to generate the sound outputs when the electrified vehicle is in the corresponding state of the vehicle drive cycle.
2. The vehicle audio control device according to claim 1, wherein the state of the electric vehicle drive cycle includes any one or more of a vehicle ON event, a vehicle OFF event, a start event, a stop event and a vehicle drive event.
3. The vehicle audio control device according to claim 2, wherein the vehicle drive event includes any one or more of an acceleration event, a deceleration event, a reverse event, a turning event, a lane change event, a cruising event and an idle event.
4. The vehicle audio control device according to claim 2, wherein the electronic controller is programmed to control the at least one speaker to change the sound output that is generated based on a change in the state of the electric vehicle drive cycle.
5. The vehicle audio control device according to claim 4, further comprising the electrified vehicle.
6. The vehicle audio control device according to claim 5, wherein the audiovisual system includes at least one electronic display programmed to display a selection of the sound outputs with respect

to the one or more states of the vehicle drive cycle.

7. The vehicle audio control device according to claim 6, wherein the electronic controller is programmed create a plurality of sound profiles, each of the sound profiles corresponding with one or more states of the vehicle drive cycle.

8. The vehicle audio control device according to claim 3, wherein the electronic controller is programmed to control the at least one speaker to change the sound output that is generated based on a change in the state of the electric vehicle drive cycle.

9. The vehicle audio control device according to claim 8, further comprising the electrified vehicle.

10. The vehicle audio control device according to claim 9, wherein the audiovisual system includes at least one electronic display programmed to display a selection for sound outputs with respect to the one or more states of the vehicle drive cycle.

11. The vehicle audio control device according to claim 10, wherein the electronic controller is programmed create a plurality of sound profiles, each of the sound profiles corresponding with one or more states of the vehicle drive cycle.
